1D SDC as a Layered DAG

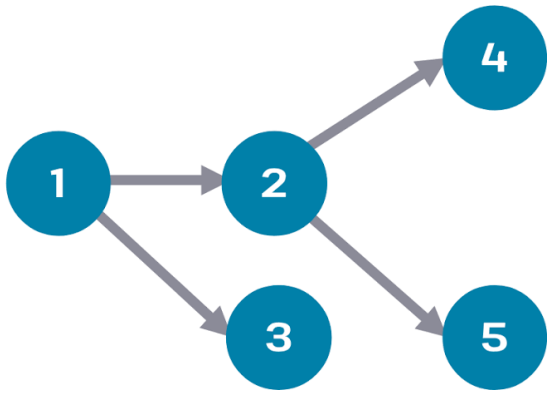
Jerry Kiely

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DAG

What is a DAG?

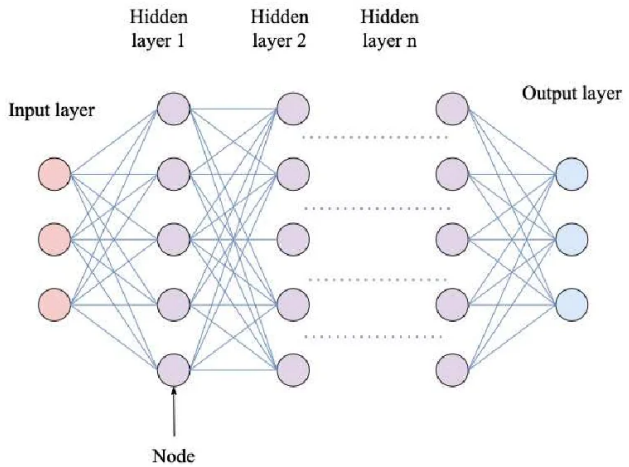
- Nodes
- Directed Edges
- No Cycles



Layered DAG

What is a Layered DAG?

- Directed Acyclic Graph
- Partitioned into Layers
- Topological ordering



Features of a Layered DAG

Why is a Layered DAG relevant to 1D SDC?

- Competing tiles occupy Layers (L_i)
- Layers are ordered ($L_i < L_j$ when $i < j$)
- Edges connect tiles from layer i to layer $i + 1$
- Edge weights - $\Delta G^{\circ}(T, r_i, l_{i+1})$

MFE and Thermodynamic Favourability

Minimum Free Energy of a 1D SDC - Dynamic Programming

- N positions
- $\leq k$ tiles per position
- $\mathcal{O}(Nk^2)$

Optimal configuration has the MFE

MFE and Thermodynamic Favourability

Tileset represented as a Layered DAG

The Shortest Path

- The Minimum Free Energy
- The most Thermodynamically Favourable Configuration

Traversal

The Shortest Path through a Layered DAG

- Topological Sort
- Edge Relaxation

```
for l in range(len(self.tiles) - 1):
    for u in self.get_layer(l):
        for v in self.get_layer(l + 1):
            uv = self.get_edge_weight(u, v, T)
            # relax edge u -> v
            if dist[v] > dist[u] + uv:
                dist[v] = dist[u] + uv
                pred[v] = u
```

Note: Layering induces a Topological Ordering

Existing MFE Implementation

How does this approach compare to other approaches?

	Time Complexity	Space Complexity
DP	$\mathcal{O}(Nk^2)$	$\mathcal{O}(k)$
Shortest Path	$\mathcal{O}(V+E)$	$\mathcal{O}(V+E)$

Performance

Next steps?

- 1D SDC implementation of an MLP
- Something

Conclusion

Questions?